

10	Cloud Logging, Monitoring, and Intrusion Detection
11	Secure API Gateway and OAuth-based Authentication
12	Compliance and Security Auditing (ISO 27001, GDPR,

E. Self-Study

Sr. No.	Topic	Unit
1	Cloud shared responsibility model – comparative study (AWS, Azure, GCP)	Unit III
2	Zero-Trust Architecture for cloud environments	Unit V
3	Cloud security threat modeling using STRIDE	Unit IV & V
4	Secure DevSecOps pipelines in cloud	All Units

Blockchain and Applications**A. OVERVIEW OF THE COURSE**

Name of Course	Blockchain and Applications
Offering Department	COMPUTER SCIENCE & ENGINEERING

Overview:

This course provides a systematic understanding of distributed ledger technology, bridging the gap between theoretical consensus models and practical enterprise applications. It covers the physics of the blockchain ecosystem, from the CAP theorem and Abstract Models (Garay, RLA) to the operational mechanics of Bitcoin and Ethereum. The course places strong emphasis on Enterprise Blockchains (Hyperledger, Corda), Smart Contract Security, and advanced cryptographic techniques like Zero-Knowledge Proofs (SNARKs) and Homomorphic Encryption. By the end, students will be capable of architecting decentralized solutions, implementing smart contracts, and analyzing the scalability and security of blockchain systems.

B. Syllabus

COURSE NO	TITLE OF THE COURSE	COURSE STRUCTURE	PRE-REQUISITE
CYCSE 105	Blockchain and Applications	3L - 0T - 2P	Cryptography and Network Security
COURSE OUTCOMES (COs) After completion of this course, the students are expected to be able to demonstrate the following knowledge, skills, and attitudes: CO1: Understand the theoretical foundations of consensus, CAP theorem, and abstract blockchain models. CO2: Analyze cryptographic primitives (ECC, Homomorphic Encryption) and Bitcoin's architectural properties. CO3: Develop and deploy secure Smart Contracts on Ethereum and understand DAO governance. CO4: Evaluate Enterprise Blockchains (Hyperledger, Corda) and their specific use cases in industry. CO5: Investigate advanced topics including Scalability, Zero-Knowledge Proofs, and Tokenomics			
COURSE CONTENTS: Unit I : Introduction & Consensus Models: History of Blockchain, Distributed Systems Basics, CAP Theorem, Byzantine Generals Problem. The Consensus Problem: Asynchronous Byzantine Agreement - AAP protocol. Nakamoto Consensus. Abstract Models: GARAY model, RLA Model. Consensus Algorithms: Proof of Work (PoW) Analysis (Consistency, Liveness, Fairness), Proof of Stake (PoS), DPoS, PoA, PoET, Paxos, Raft. Unit II: Cryptography & Bitcoin Architecture: Cryptographic Basics: Hashing (SHA-256), Digital Signatures, Elliptic Curve Cryptography (ECC), Homomorphic Encryption, Bitcoin Mechanics: Blocks, Transactions, Merkle Trees, Hardness of Mining, Forks, Double Spending, Mathematical analysis of Bitcoin properties. Anonymity and UTXO model. Unit III: Ethereum & Smart Contracts: Ethereum Virtual Machine (EVM), Gas Mechanism, Accounts (EOA vs Contract), Smart Contracts: Introduction to Solidity, Ricardian Contracts, Smart Contract Templates, Oracles. Attacks on Smart Contracts: Reentrancy, Overflow, DAO attacks. DAOs: Governance and Tools (Aragon, DAOstack).			

Approved by BoS held on 18/02/2026

Unit IV: Enterprise Blockchains & Alternatives:

Hyperledger: Fabric Architecture, Composer, Chaincode, MSP, Channels, R3 Corda: Architecture, States, Vaults, Notaries, CorDapps, Alternative Chains: Ripple (Consensus), Quorum, BigchainDB. Permissioned vs. Permissionless systems

Unit V: Advanced Trends, Scalability & Privacy: **Scalability:** The Scalability Trilemma, Layer 2 Solutions (Sidechains, Payment Channels/Lightning Network), Privacy & Security: Zero Knowledge Proofs (SNARKs, Zcash), Pairing on Elliptic Curves. Token Economy: Tokenomics, ICOs, STOs. Research: Blockchain in IoT/VANET

SUGGESTED READINGS

1. Arvind Narayanan, Joseph Bonneau, et al. *Bitcoin and Cryptocurrency Technologies: A Comprehensive Introduction*. Princeton University Press, 2016.
2. Jonathan Katz and Yehuda Lindell. *Introduction to Modern Cryptography, Second Edition*. CRC Press.
3. Ritesh Modi. *Solidity Programming Essentials*. Packt Publishing.
4. Wattenhofer, R. P. *Distributed Ledger Technology: The Science of the Blockchain*. Inverted Forest Publishing

C. Theory Lecture Plan

S. No.	CONTENT	No. of Lectures	Unit
1	History of Blockchain, Distributed Systems Basics, CAP Theorem, Byzantine Generals Problem. The Consensus Problem: Asynchronous Byzantine Agreement - AAP protocol. Nakamoto Consensus	4	Unit I
2	Consensus Problem: Asynchronous Byzantine Agreement - AAP protocol. Nakamoto Consensus. Abstract Models: GARAY model, RLA Model. Consensus Algorithms: Proof of Work (PoW) Analysis (Consistency, Liveness, Fairness), Proof of Stake (PoS), DPoS, PoA, PoET, Paxos, Raft	4	Unit I
3	Cryptographic Basics: Hashing (SHA-256), Digital Signatures, Elliptic Curve Cryptography (ECC), Homomorphic Encryption	4	Unit II
4	Bitcoin Mechanics: Blocks, Transactions, Merkle Trees, Hardness of Mining, Forks, Double Spending, Mathematical analysis of Bitcoin properties. Anonymity and UTXO model	4	Unit II
Class Test – I			
5	Ethereum Virtual Machine (EVM), Gas Mechanism, Accounts (EOA vs Contract), Smart Contracts: Introduction to Solidity, Ricardian Contracts, Smart Contract Templates, Oracles	3	Unit III
6	Attacks on Smart Contracts: Reentrancy, Overflow, DAO attacks. DAOs: Governance and Tools (Aragon, DAOstack)	4	Unit III
Mid-Semester Evaluation			
7	Hyperledger: Fabric Architecture, Composer, Chaincode, MSP, Channels, R3 Corda: Architecture, States, Vaults, Notaries, CorDapps	4	Unit IV
8	Alternative Chains: Ripple (Consensus), Quorum, BigchainDB. Permissioned vs. Permissionless systems.	4	Unit IV
9	Advanced Trends, Scalability & Privacy: Scalability: The Scalability Trilemma, Layer 2 Solutions (Sidechains, Payment Channels/Lightning Network).	4	Unit V
10	Privacy & Security: Zero Knowledge Proofs (SNARKs, Zcash), Pairing on Elliptic Curves. Token Economy: Tokenomics, ICOs, STOs. Research: Blockchain in IoT/VANET.	3	Unit V

D. Lab Experiments

No.	Name of Experiment
1	Blockchain Simulation: Implement Merkle Tree and Blockchain Hash function in Python.

Approved by BoS held on 18/02/2026

2	Mining Implementation: Simulate Proof of Work (Mining) and create a basic Cryptocurrency in Python.
3	Consensus Logic: Design and implement PoS/PoW Consensus Algorithm using Python.
4	Ethereum Setup: Set up Environment (Windows/Ubuntu), Install Wallet, Send/Receive Ether.
5	Smart Contracts: Create, Deploy, and Interact with Smart Contracts using Solidity (Remix/Truffle).
6	DeFi Application: Implement a "Bank" Contract (Deposit/Withdraw) and connect to a Web Frontend.
7	Enterprise Lab: Setup Hyperledger Fabric Network and deploy a basic Chaincode (Smart Contract).

E. Self-Study

Sr. No.	Topic	Unit
1	Blockchain-based secure data sharing system	Unit III
2	Supply chain traceability using blockchain	Unit V
3	Healthcare record management on blockchain	Unit IV & V
4	IoT device authentication using blockchain	All Units

Digital Forensic

A. Overview

Digital Forensics is a specialized domain of cyber security focused on the systematic identification, acquisition, preservation, examination, and presentation of digital evidence in a legally admissible manner. It involves forensic imaging of storage media, analysis of file systems, recovery of deleted data, examination of volatile memory, network traffic inspection using tools such as Wireshark, and artifact analysis from operating systems, browsers, emails, and mobile devices. The discipline follows strict chain-of-custody procedures and employs cryptographic hashing (e.g., MD5, SHA-256) to ensure evidence integrity. Digital forensics supports incident response, cybercrime investigation, corporate fraud detection, and legal proceedings by reconstructing events and attributing actions through scientifically validated methodologies.

Course No.	Title of the Course	Credits	Course Structure	Pre-Requisite
CYCSE 106	Digital Forensics	4	3L-0T-2P	
COURSE OUTCOMES: CO1: To understand the fundamentals of digital forensics and different kinds of cyber-crimes. CO2: To apply different types of data acquisition tools CO3: To analyze the digital investigation scene CO4: To learn about the digital forensics software and hardware tools CO5: To understand the virtual machine and network forensics				

B. Course Contents

Unit No.	Topics
Unit 1	Introduction: Digital Forensics Fundamentals, Computer Crimes, Understanding Digital Investigations, Procedures for Private-Sector High-Tech Investigations, Understanding Data Recovery Workstation and Software, Conducting an Investigation, Investigator's Office and Laboratory
Unit 2	Data Acquisition: Understanding Storage Formats for Digital Evidence, Determining the Best Acquisition Method, Acquisition tools, Validating Data Acquisitions, Performing RAID Data Acquisitions, Remote Network Acquisition Tools, Other Forensics Acquisition Tools
Unit 3	Processing Crimes and Incident Scenes: Identifying Digital Evidence, Collecting Evidence in Private-Sector Incident Scenes, Processing Law Enforcement Crime Scenes, Preparing for a Search, Securing a Digital Incident or Crime

Approved by BoS held on 18/02/2026